

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Ethidium bromide, 1% Solution/Molecular Biology

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>BP1302-10</b>                                                                                     |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

Acute Inhalation Toxicity - Dusts and Mists  
Germ Cell Mutagenicity

Category 3  
Category 2

#### Environmental hazards

No hazards identified

### Label Elements



Skull and Crossbones



Health Hazard

Signal Word

Danger

**Hazard Statements**

H331 - Toxic if inhaled

H341 - Suspected of causing genetic defects if inhaled

**Precautionary Statements**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P311 - Call a POISON CENTER or doctor

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P405 - Store locked up

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

Other information

## Section 3 - Composition and Information on Ingredients

| Component                                           | CAS No    | Weight % |
|-----------------------------------------------------|-----------|----------|
| Water                                               | 7732-18-5 | 99       |
| 3,8-Diamino-1-ethyl-6-phenylphenantridinium bromide | 1239-45-8 | 1        |

## Section 4 - First Aid Measures

|                                     |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| Ingestion                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Skin Contact                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Eye Contact                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| General Advice                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| Self-Protection of the First Aider  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |
| First Aid Facilities                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                           |
| Most important symptoms and effects | None reasonably foreseeable.                                                                                                                                                                                                                                                                                                   |
| Notes to Physician                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                         |

## Section 5 - Fire Fighting Measures

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

#### **Hazardous Decomposition Products**

Nitrogen oxides (NO<sub>x</sub>).

#### **Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

#### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **Section 6 - Accidental Release Measures**

#### **Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

#### **Environmental Precautions**

Should not be released into the environment.

#### **Methods for Containment and Clean Up**

##### **Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

##### **Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

#### **Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

#### **Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a cool, well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## **Section 8 - Exposure Controls and Personal Protection**

#### **Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Natural rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |                 |                       |
| Neoprene       |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

#### Recommended Filter type:

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)

#### Recommended half mask:-

Particle filtering: EN149:2001 (or AUS/NZ equivalent)

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

### Environmental exposure controls

No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

#### Appearance

Red

#### Physical State

Liquid

#### Odor

Odorless

#### Odor Threshold

No data available

#### pH

No information available

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

No information available

#### Flash Point

No data available

Method - No information available

|                                                   |                          |             |
|---------------------------------------------------|--------------------------|-------------|
| <b>Evaporation Rate</b>                           | negligible               |             |
| <b>Flammability (solid,gas)</b>                   | Not applicable           | Liquid      |
| <b>Explosion Limits</b>                           | No data available        |             |
| <b>Vapor Pressure</b>                             | negligible               |             |
| <b>Vapor Density</b>                              | No data available        | (Air = 1.0) |
| <b>Specific Gravity / Density</b>                 | No data available        |             |
| <b>Bulk Density</b>                               | Not applicable           | Liquid      |
| <b>Water Solubility</b>                           | Miscible                 |             |
| <b>Solubility in other solvents</b>               | No information available |             |
| <b>Partition Coefficient (n-octanol/water)</b>    |                          |             |
| <b>Component</b>                                  | <b>log Pow</b>           |             |
| 3,8-Diamino-1-ethyl-6-phenylphenantridium bromide | -0.38                    |             |
| <b>Autoignition Temperature</b>                   | No data available        |             |
| <b>Decomposition Temperature</b>                  | No data available        |             |
| <b>Viscosity</b>                                  | No data available        |             |
| <b>Explosive Properties</b>                       | No information available |             |
| <b>Oxidizing Properties</b>                       | No information available |             |

Other information

## Section 10 - Stability and Reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat.        |
| <b>Incompatible Materials</b>           | Strong oxidizing agents.                   |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx).                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.   |

## Section 11 - Toxicological Information

Information on Toxicological Effects

Product Information

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Based on available data, the classification criteria are not met |
| <b>Dermal</b>              | No data available                                                |
| <b>Inhalation</b>          | Category 3                                                       |

Toxicology data for the components

| Component                                         | LD50 Oral        | LD50 Dermal               | LC50 Inhalation |
|---------------------------------------------------|------------------|---------------------------|-----------------|
| Water                                             | -                | -                         | -               |
| 3,8-Diamino-1-ethyl-6-phenylphenantridium bromide | 1503 mg/kg (rat) | LD50 > 2000 mg/kg ( Rat ) |                 |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

**(e) germ cell mutagenicity;** Category 2

Possible risk of irreversible effects; Mutagenic effects have occurred in experimental animals

**(f) carcinogenicity;** No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

|               |                           |
|---------------|---------------------------|
| Target Organs | No information available. |
|---------------|---------------------------|

**(j) aspiration hazard;** No data available

**Symptoms / effects, both acute and delayed** No information available

## Section 12 - Ecological Information

|                            |                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b> | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|

|                                      |                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------|
| <b>Persistence and Degradability</b> |                                                                               |
| <b>Persistence</b>                   | Miscible with water, Persistence is unlikely, based on information available. |
| <b>Bioaccumulative Potential</b>     | Bioaccumulation is unlikely                                                   |

| Component                                           | log Pow | Bioconcentration factor (BCF) |
|-----------------------------------------------------|---------|-------------------------------|
| 3,8-Diamino-1-ethyl-6-phenylphenantridinium bromide | -0.38   | No data available             |

|                 |                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mobility</b> | The product is water soluble, and may spread in water systems. : Will likely be mobile in the environment due to its water solubility Highly mobile in soils |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors |
|----------------------------------------|---------------------------------------------------------------------------|

|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| <b>Persistent Organic Pollutant</b> | This product does not contain any known or suspected substance |
|-------------------------------------|----------------------------------------------------------------|

|                                  |                                                                |
|----------------------------------|----------------------------------------------------------------|
| <b>Ozone Depletion Potential</b> | This product does not contain any known or suspected substance |
|----------------------------------|----------------------------------------------------------------|

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                               |                                                                           |
|-------------------------------|---------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point. |
|-------------------------------|---------------------------------------------------------------------------|

|                          |                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Other Information</b> | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Section 14 - Transport Information

**IMDG/IMO**

|                         |                               |
|-------------------------|-------------------------------|
| UN-No                   | UN2810                        |
| Proper Shipping Name    | Toxic liquid, organic, n.o.s. |
| Technical Shipping Name | Ethidium bromide solution     |
| Hazard Class            | 6.1                           |
| Packing Group           | III                           |

**ADG**

|                         |                               |
|-------------------------|-------------------------------|
| UN-No                   | UN2810                        |
| Proper Shipping Name    | Toxic liquid, organic, n.o.s. |
| Technical Shipping Name | Ethidium bromide solution     |
| Hazard Class            | 6.1                           |
| Packing Group           | III                           |

**IATA**

|                         |                               |
|-------------------------|-------------------------------|
| UN-No                   | UN2810                        |
| Proper Shipping Name    | Toxic liquid, organic, n.o.s. |
| Technical Shipping Name | Ethidium bromide solution     |
| Hazard Class            | 6.1                           |
| Packing Group           | III                           |

|                        |                                 |
|------------------------|---------------------------------|
| Environmental hazards  | No hazards identified           |
| Special Precautions    | No special precautions required |
| Additional information | None known                      |

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National Regulations**                      **Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                                                           | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3,8-Diamino-1-ethyl-6-phenylphenantridiniu<br>m bromide - 1239-45-8 | Schedule 5 listed - in preparations except: a) when separately specified in these Schedules, b) Dialkyl or dialkoyl quaternary ammonium compounds where the alkyl or alkoyl groups are derived from tallow or hydrogenated tallow or similar chain length [C16/C18] source, or c) in preparations containing <=5% of such Quaternary ammonium compounds<br>Schedule 6 listed - except: a) when separately specified in these Schedules, b) when included in Schedule 5, c) Dialkyl or Dialkoyl quaternary ammonium compounds where the Alkyl or Alkoyl groups are derived from Tallow or hydrogenated Tallow or similar chain length [C16/C18] sources, or d) in preparations containing <=5% of such Quaternary ammonium compounds |

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component         | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-------------------|-------------------------------------------------------------|------------------------|
| Water - 7732-18-5 | Present                                                     | -                      |

|                                                                 |         |   |
|-----------------------------------------------------------------|---------|---|
| 3,8-Diamino-1-ethyl-6-phenylphenantridinium bromide - 1239-45-8 | Present | - |
|-----------------------------------------------------------------|---------|---|

#### Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

National pollutant inventory Not applicable

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### International Inventories

| Component                                           | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------------------------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Water                                               | X    | X     | 231-791-2 | -      | X    | X   | -    | X     | X    |      | X     | KE-35400 |
| 3,8-Diamino-1-ethyl-6-phenylphenantridinium bromide | X    | X     | 214-984-6 | -      | -    | -   | -    | X     | -    |      | X     | KE-09740 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

#### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Not applicable.

| Component                                           | CAS No    | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------------------------------------|-----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Water                                               | 7732-18-5 | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| 3,8-Diamino-1-ethyl-6-phenylphenantridinium bromide | 1239-45-8 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

#### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous | REACH Regulation (EC 1907/2006) article 59 - Candidate |
|-----------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
|-----------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|



|                                                             | Authorization | Substances                                                         | List of Substances of Very High Concern (SVHC) |
|-------------------------------------------------------------|---------------|--------------------------------------------------------------------|------------------------------------------------|
| 3,8-Diamino-1-ethyl-6-phenylpiperidine nantridinium bromide | -             | Use restricted. See item 75.<br>(see link for restriction details) | -                                              |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances<br><b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory<br><b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List<br><b>IECSC</b> - Chinese Inventory of Existing Chemical Substances<br><b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances<br><b>TWA</b> - Time Weighted Average<br><b>IARC</b> - International Agency for Research on Cancer<br><b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association<br><b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships<br><b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land<br><b>LD50</b> - Lethal Dose 50%<br><b>EC50</b> - Effective Concentration 50%<br><b>WEL</b> - Workplace Exposure Limit<br><b>DNEL</b> - Derived No Effect Level<br><b>POW</b> - Partition coefficient Octanol:Water<br><b>vPvB</b> - very Persistent, very Bioaccumulative<br><b>VOC</b> - (Volatile Organic Compound) | <b>NZIoC</b> - New Zealand Inventory of Chemicals<br><b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances<br><b>ENCS</b> - Japanese Existing and New Chemical Substances<br><br><b>KECL</b> - Korean Existing and Evaluated Chemical Substances<br><b>CAS</b> - Chemical Abstracts Service<br><b>ACGIH</b> - American Conference of Governmental Industrial Hygienists<br>Predicted No Effect Concentration (PNEC)<br><b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code<br><b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail<br><b>OECD</b> - Organisation for Economic Co-operation and Development<br><b>LC50</b> - Lethal Concentration 50%<br><b>ATE</b> - Acute Toxicity Estimate<br><b>RPE</b> - Respiratory Protective Equipment<br><b>NOEC</b> - No Observed Effect Concentration<br><b>BCF</b> - Bioconcentration factor<br><b>PBT</b> - Persistent, Bioaccumulative, Toxic |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.

|                         |                 |
|-------------------------|-----------------|
| <b>Revision Date</b>    | 21-Nov-2022     |
| <b>Revision Summary</b> | Not applicable. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

---

**End of Safety Data Sheet**